PRACTICAL NO 4

Aim: Find a procedure to transfer the files from one virtual machine to another virtual machine

Procedure:

Step 1 : Install Virtual Box https://www.virtualbox.org/ Click on download and select windows host.

Step 2 : Download VM image

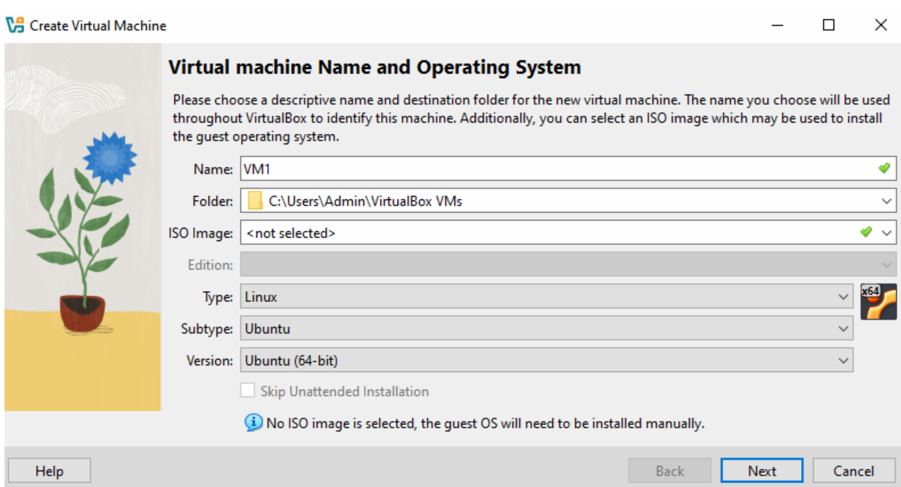
https://drive.google.com/drive/u/0/folders/1me_nJJh0fvdDOXX3ew2jzGQpoP7f_iFt

Step 3 : Open Virtual Box

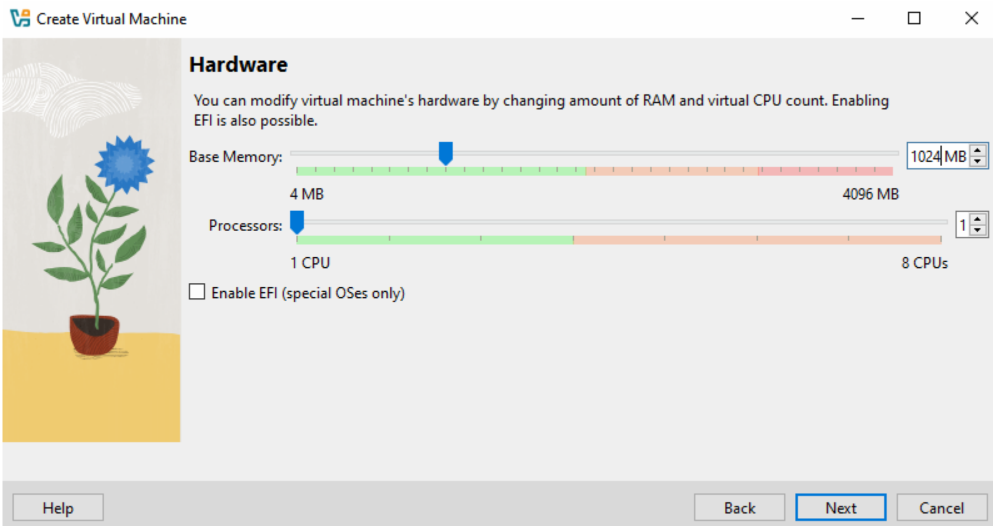
Step 4 : Create Virtual Machines (We are creating 2 VMs)

How to create VM:

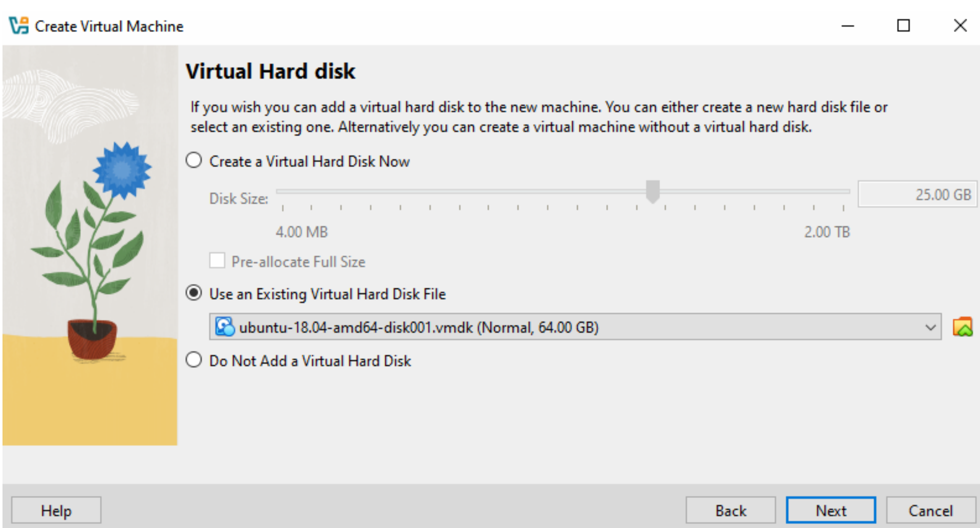
1. Click on New – Give Name to VM – Select type Linux – Select subtype Ubuntu – Version Ubuntu (64 Bit). Click on Next



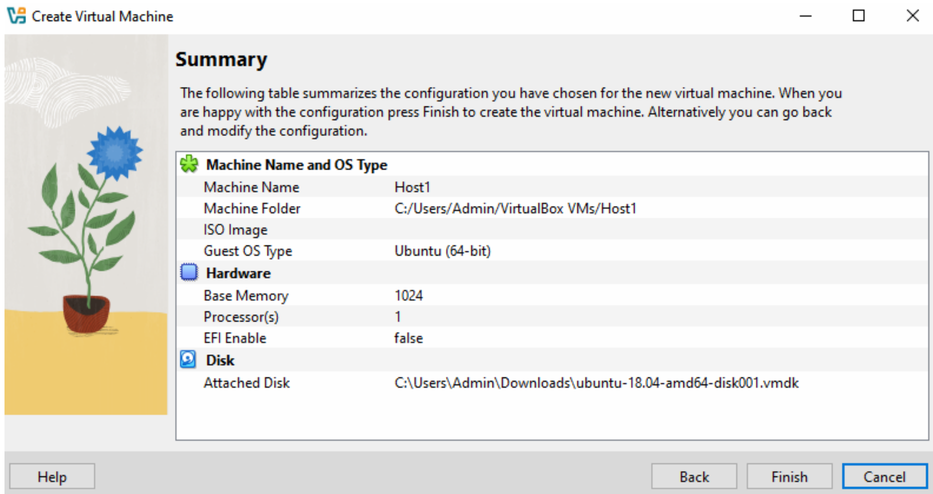
2. On base memory resize it as 1024 MB. Click on Next.



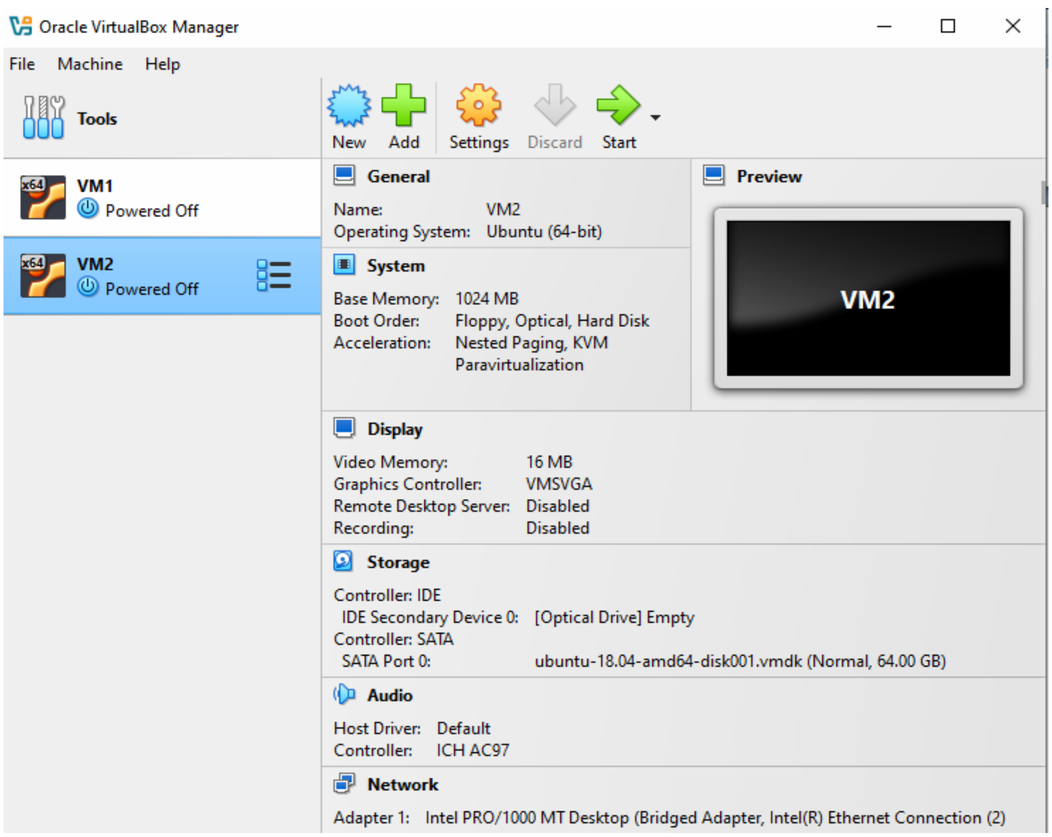
3. Select Use an Existing Virtual Hard Disk File- Select the image by clicking on folder symbol – click on add – then select the image that we have downloaded through the drive link. Click on Next.



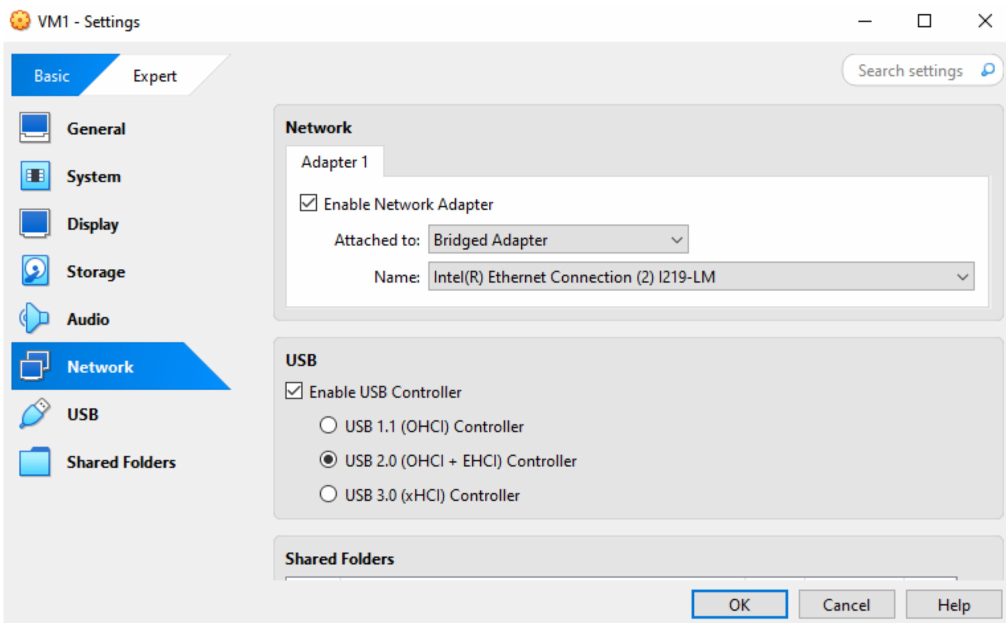
4. We will be on final stage, it will show you the summery.



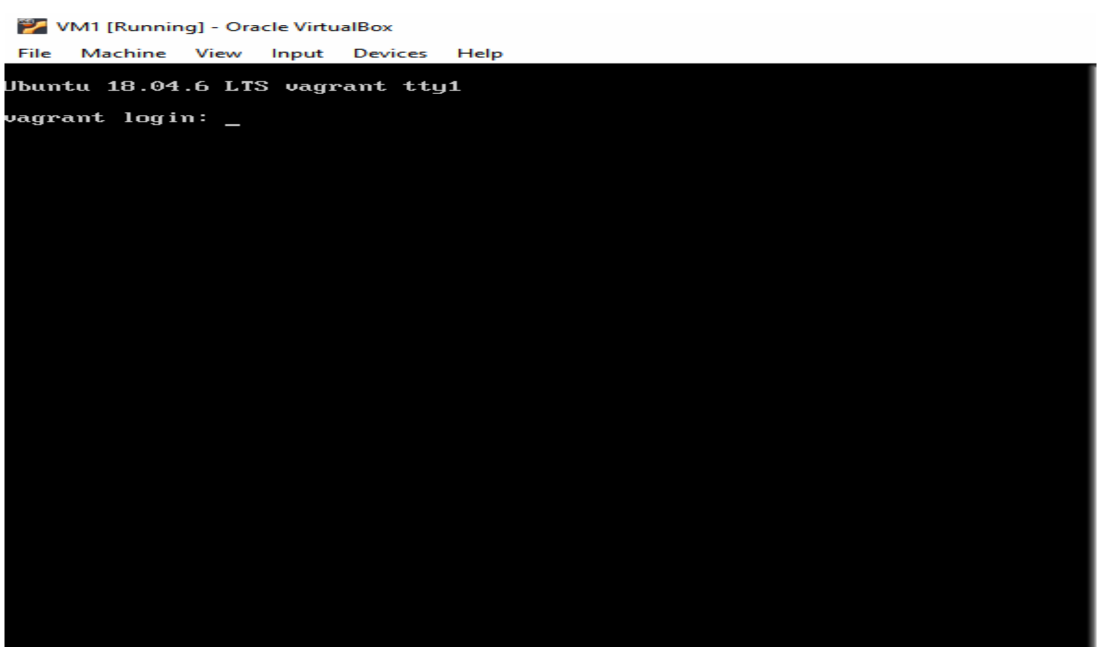
5. In this way we created one VM by using same way we have to create second VM.



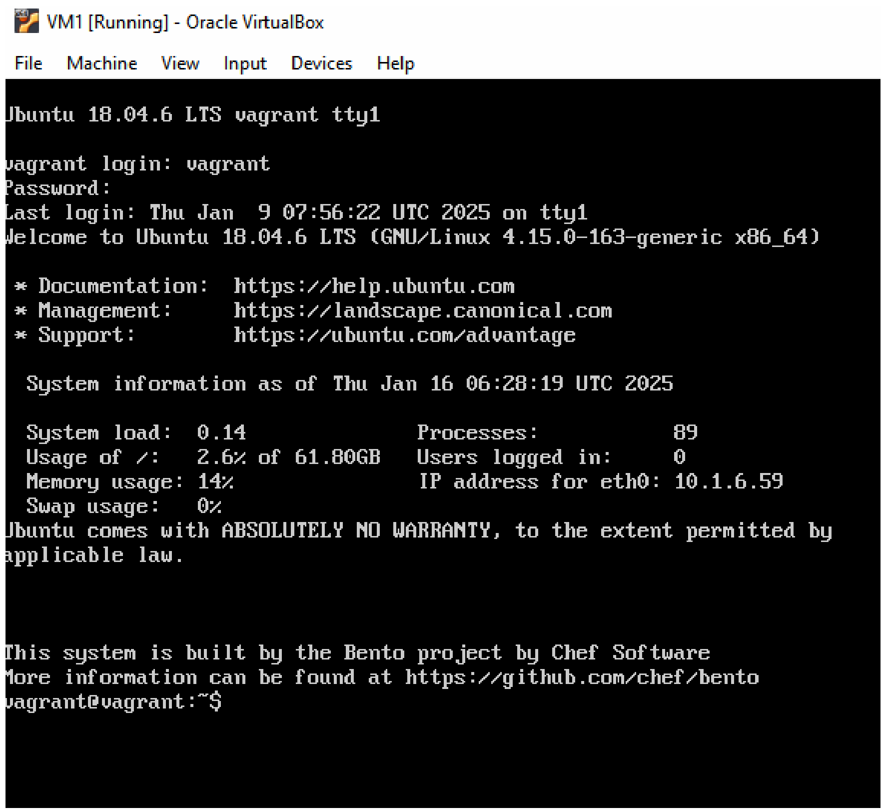
6. One more setting we need to give – select VM1 – Go to setting – click on network – select Bridge network rather than NAT network – click on OK.
Same setting need to do for VM2.



Step 5 : Boot the VM by double clicking it or select VM1 and click on start. (Wait for booting the VM – It will take time.)



Step 6 : By using same way boot second VM. (Login ID and Password for VMs will be **vagrant**)



Step 7 : There are some command that we are using in linux by using commands we are going to transfer file from one VM to another VM.

Some commands are listed out below:

ls - list all the files and directories

cat - show the content inside a file

scp - it will help to copy files form one vm to other

cd - change directoory

mkdir - make a new directory

touch - it makes a new file

nano - nano is a editor inside linux OS

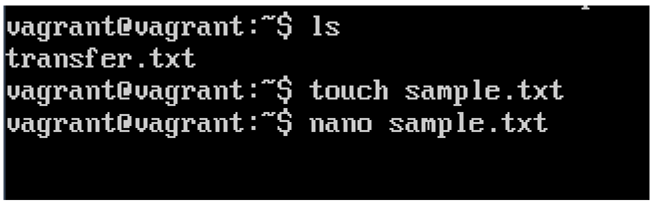
Firstly we will use ls command for checking that is there any file or not.

Step 8 : Create one file by using touch command. (here I am using sample as a file name.)

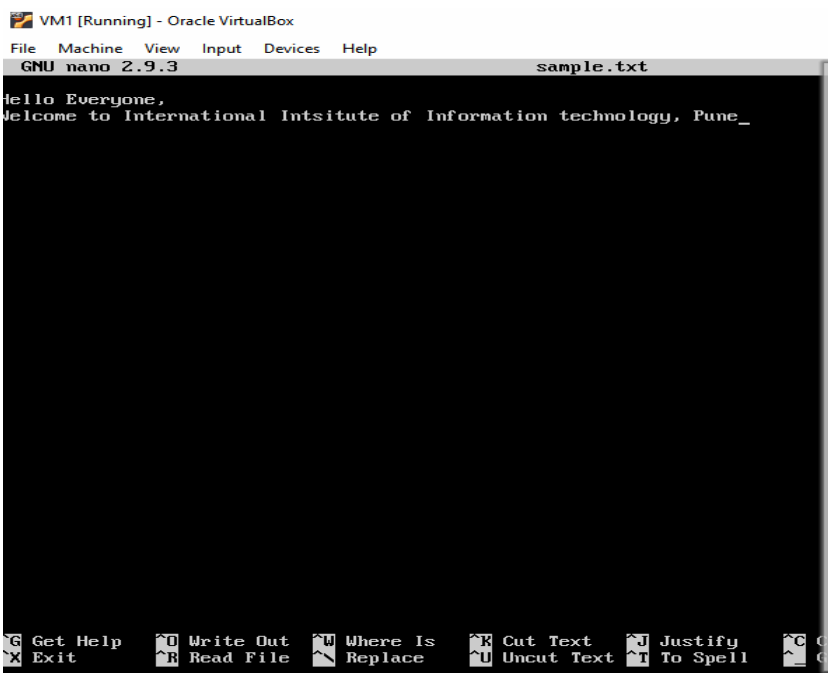
Example : touch sample.txt

Step 9 : Now add some content in file by using nano command.

Example : nano sample.txt



Step 10 : Add content – Press Ctr+X – then press Y - then enter



Check the file is added to directory or not by using ls commnd.

You can check the content of your file by using cat command

Example – cat sample.txt

Step 11: Now we have created a file, We will send this file to our second VM (VM2).

For that purpose we should know the IP address of Secondary VM where we want to send a file.

So open the second VM by using same procedure – give id and password (same password required) and give the command as ifconfig – you will get the IP address. As per below image 10.1.6.250 is an IP address that we are using for transfer the file.

```
vagrant@vagrant:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.1.6.250  netmask 255.255.0.0  broadcast 10.1.255.255
        inet6 fe80::a00:27ff:fe3d:4934  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:3d:49:34  txqueuelen 1000  (Ethernet)
        RX packets 2960  bytes 488269 (488.2 KB)
        RX errors 0  dropped 44  overruns 0  frame 0
        TX packets 91  bytes 8232 (8.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 936 (936.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 936 (936.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Step 12: Now we know the IP address of VM2 – Go back to VM1

For transfer the file from one VM to another we need scp command.

Example – scp sample.text vagrant@10.1.6.250:/home/vagrant

It will ask you the password, so again same password we have to use i.e. vagrant.

File get transfer – refer below image.

```
vagrant@vagrant:~$ scp sample.txt vagrant@10.1.6.250:/home/vagrant
vagrant@10.1.6.250's password:
sample.txt                                                      100%   83     53.5KB/s   00:00
vagrant@vagrant:~$
```

Step 13 : File is transferred from one VM, Now we will check it on another VM that its received or not by using ls command.

```
vagrant@vagrant:~$ ls
sample.txt  transfer.txt
vagrant@vagrant:~$
```

Our file is transferred from one VM to another.

```
vagrant@vagrant:~$ ls
sample.txt  transfer.txt
vagrant@vagrant:~$ cat sample.txt
Hello Everyone,
Welcome to International Intsitute of Information technology, Pune
vagrant@vagrant:~$
```

You can check the content of file again.

In this way we have transfer a file using virtual machine and using different OS.

--------------------------------------In Case of Error--

In case you get an error while booting the VM (if by default virtualization and hypervior is enable in your system then you will not get an error)

(If virtualization is disable then it is the possibility that you will get error while booting the VM.)

Error will be like- **error -vt x is disabled in the bios for all cpu modes (all-vmx-disabled)**

Solution for this Error:

Go to system setting – go to Update & Security – Go to Recovery – In advanced startup click on restart now

You will see a blue screen click on troubleshoot – click on UEFI Firmware Setting – Restart Now

Go to System – then System Security – Enable virtualization technology and virtualization technology directed I/O

Save changes and exit from it. Your system will get restart. Then reopen virtual box and boot your VM it will get start.